Diego De Pablo

Software Engineer | Backend, Data & Applied AI

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Professional Summary

Software Engineer and Bioinformatics graduate specializing in backend development, data engineering, and applied artificial intelligence. Proven track record architecting modular microservices, concurrent data ingestion pipelines, and hybrid semantic search engines (dense vector + BM25 lexical) powered by schema-validated AI agents. Graduated with top academic honours (multiple Matrículas de Honor) and currently pursuing an M.S. in Computer Engineering.

Professional Experience

Software Engineer, Khaos Research — **Sept 2025 – Present**

European federated research data platform engineering (SEDIA | Gaia-X)

- Engineered high-throughput backend services and integration modules for the European **SEDIA** project, adhering to federated and interoperable data space standards under **Gaia-X** specifications.
- Architected and integrated a hybrid search engine combining dense vector retrieval in **Qdrant** with BM25 lexical ranking in **Elasticsearch**, orchestrated via agents validated with strict **Pydantic AI** contracts.
- Developed the concurrent **MetaDataXtract** pipeline for automated ingestion and metadata extraction across +30 scientific data formats, incorporating SHA-256 integrity validation and automated VirusTotal API auditing.
- Optimized data ingestion throughput and orchestrated containerized **Docker** deployments with structured observability and error tracing.

Full-Stack Developer, mSurgery — **March 2025 – July 2025**

Medical telemetry platforms and real-time surgical digital twins

- Led the frontend migration to **Svelte 5** adopting the reactive *runes* paradigm, decoupling application state to ensure high frame rates within the real-time 3D surgical viewport.
- Built interactive clinical interfaces for the **SMOTTS** medical simulation platform, synchronizing real-time digital twin telemetry for procedural accuracy.
- Delivered technical demonstrations and platform presentations in English to international investors and healthcare review committees at the National Digital Content Hub.

Junior Software Engineer, T&B Company Group — **June 2021 – December 2021**

Enterprise platform evolution and REST microservice modernization

- Refactored legacy enterprise modules in **Java** and **Angular** toward decoupled architectures, resolving critical production issues and improving codebase testability.
- Defined standardized API interface contracts using **OpenAPI / Swagger** and documented REST microservices to accelerate cross-team frontend/backend integration.

Featured Technical Projects

MetaDataSearch: AI Semantic Search & Hybrid Filtering

Stack: Next.js, FastAPI, Qdrant (Vector DB), Elasticsearch, Pydantic AI, Docker

- Hybrid search engine tailored for scientific dataset repositories: couples dense vector retrieval with BM25 text ranking, parsing natural language queries into structured faceted filters automatically.
- Containerized microservice architecture with asynchronous, low-latency FastAPI endpoints.

Biomedical Telemonitoring IoT Smartband (Bachelor's Thesis)

Stack: C++, FreeRTOS, ESP32, MAX30102, MPU-6050, MQTT, FastAPI, TimescaleDB, Docker

- End-to-end telemonitoring system: multi-threaded ESP32 firmware for vital signs acquisition via photoplethysmography (pulse and SpO2) and fall detection algorithm using MPU-6050 accelerometer data.
- Real-time MQTT telemetry synchronization to a FastAPI backend and TimescaleDB time-series database, coupled with a clinician web dashboard. **Grade: 10/10 with Highest Academic Distinction (Matrícula de Honor)**.

TITAN Workflow: Scientific Workflow Composition Assistant

Stack: FastAPI, Pydantic AI, LangChain, LLMs / Embeddings, Docker

- AI-assisted composition engine for computational scientific workflows: validates data schemas, dependency graphs, and input/output contracts in Pydantic AI prior to task orchestration.

Cloud-Native Platform & GitOps Pipeline

Stack: Kubernetes, Helm, Docker, Argo CD, Prometheus, Grafana, GitHub Actions

- Automated infrastructure for deploying scalable microservices and search indices, featuring centralized Prometheus/Grafana monitoring, latency metrics, and GitOps-driven reconciliation via Argo CD.

Education

M.S. in Computer Engineering — **2025 – Present**

University of Málaga

- Advanced coursework in distributed computing, large-scale software architectures, data-intensive systems, and applied artificial intelligence.

B.S. in Health Engineering (Bioinformatics Specialization) — **2021 – 2025**

University of Málaga

- Graduated with top academic honours. **Highest Distinction (Matrícula de Honor)** in core disciplines: *Programming Fundamentals*, *Intelligent Systems*, and *Bioinformatics Projects*.
- Bachelor's Thesis (TFG) in Biomedical IoT & Telehealth awarded **10/10 with Highest Academic Distinction**.

Technical Skills & Languages

Programming Languages: Python, TypeScript, JavaScript, C++, SQL, Java, Bash, R.

Backend & Architecture: FastAPI, REST Microservices, OpenAPI / Swagger, Pydantic, Concurrency (AsyncIO), Node.js.

Databases & Search: PostgreSQL, Elasticsearch, Qdrant (Vector DB), TimescaleDB, Redis.

AI & Data Engineering: Pydantic AI, LangChain, RAG / Vector Embeddings, LLM Agents, PyTorch.

DevOps, Cloud & Systems: Docker, Kubernetes, Helm, Argo CD (GitOps), CI/CD (GitHub Actions), Linux, FreeRTOS (ESP32).

Frontend & UI: React, Svelte 5, Next.js, HTML5, CSS3, Tailwind CSS.

Languages: Spanish (Native) · **English** (B2 Professional / Demonstrated technical fluency in international pitches).